

UGANDA MARKET MASTER - MMIS PRODUCTION SCHEMA v5.0

Comprehensive Complete Schema Proposal

Based on latest Knowledge Base (Optimized + Recommended + Final-Optimized + All Module Enhancements)

Executive Summary

This is the **unified, production-grade schema** consolidating:

- `schema.optimized_Version2 .prisma` + `schema.recommended .prisma` + `schema.final-optimized_Version1 .prisma`
- All enhanced financial/revenue/delinquency/tax models from comparison reports
- Missing modules (Disaster & Risk, Compensation, Government Integration, IoT/CCTV, POS, HR, Parking, AI, etc.)
- Full support for 21 cities, 8 user roles, multi-market ecosystem, Uganda Data Protection Act & URA compliance

Next Steps (as per your instruction):

1. **This message** = Complete **Models + Enums + Tables + Columns** list
2. **Your feedback** on required attributes / business logic alignments
3. **Next response** = Final Prisma schema + relations + full ERD (Mermaid + PlantUML) + migration plan

1. COMPLETE ENUMS LIST (Unified & Production-Ready)

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prisma

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// CORE ENUMS - UNIFIED PRODUCTION SET
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enum UserStatus { ACTIVE INACTIVE SUSPENDED PENDING DELETED }
enum KycStatus { NOT_SUBMITTED PENDING UNDER_REVIEW VERIFIED REJECTED }
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EXPIRED }
enum AdminLevel { SUPER_ADMIN NATIONAL_ADMIN DISTRICT_ADMIN CITY_ADMIN
MARKET_MASTER PSEUDO_MARKET_ADMIN }
enum PseudoMarketRole { REVENUE_COLLECTOR HEALTH_INSPECTOR SECURITY_ADMIN
GATE_COUNTER STOCK_COUNTER VAT_PAYMENT_COUNTER WAREHOUSE_MANAGER
QUALITY_CONTROLLER }
enum StakeholderType { MARKET_AUTHORITY MEMBER VENDOR SUPPLIER CUSTOMER
GUEST }
enum MfaType { EMAIL SMS TOTP NONE }
enum VerificationTokenType { EMAIL_VERIFICATION PHONE_VERIFICATION
PASSWORD_RESET ACCOUNT_RECOVERY ADMIN_INVITATION KYC_VERIFICATION
HANDSHAKE_CODE }
enum InvitationStatus { PENDING ACCEPTED EXPIRED REVOKED USED }
enum InvitationType { USER_REGISTRATION PASSWORD_RESET ADMIN_ONBOARDING
VENDOR_REGISTRATION SUPPLIER_REGISTRATION KYC_VERIFICATION }
enum PermissionResource { SYSTEM USER MARKET CITY DISTRICT FACILITY
PRODUCT INVENTORY GATE PAYMENT TAX REPORT SETTINGS KYC INVITATION ROLE
PERMISSION DOCUMENT AUDIT NOTIFICATION PROCUREMENT DISASTER COMPENSATION
IOT }
enum PermissionAction { CREATE READ UPDATE DELETE MANAGE APPROVE VERIFY
EXPORT IMPORT ASSIGN REVOKE }
enum DocumentType { ID_PROOF ADDRESS_PROOF BUSINESS_LICENSE
TAX_REGISTRATION BANK_STATEMENT UTILITY_BILL PASSPORT DRIVING_LICENSE
VENDOR_CERTIFICATE SUPPLIER_CERTIFICATE QUALITY_CERTIFICATE
INSURANCE_POLICY OTHER INVOICE RECEIPT GATE_TOKEN FACILITY_CONTRACT
SYSTEM_REPORT }
enum MarketType { PERMANENT TEMPORARY POP_UP WEEKLY_BAZAAR SEASONAL }
enum FacilityType { SHOP STALL WAREHOUSE KIOSK OFFICE OTHER }
enum FacilityStatus { ACTIVE INACTIVE UNDER_MAINTENANCE CLOSED }
enum OccupationStatus { VACANT OCCUPIED RESERVED LOCKED }
enum DepositStatus { NOT_COLLECTED PARTIAL FULL FORFEITED REFUNDED }
enum MarketTokenType { GATE_ENTRY GATE_EXIT STOCK_RECEIPT STOCK_DISPATCH
PAYMENT_CONFIRMATION TAX_PAYMENT RENT_PAYMENT INSPECTION_APPROVAL
DELIVERY_CONFIRMATION VEHICLE_PASS VISITOR_PASS TEMPORARY_ACCESS }
enum TokenStatus { PENDING ACTIVE USED EXPIRED REVOKED INVALIDATED }
enum QrCodeType { USER_IDENTIFICATION STAFF_IDENTIFICATION VEHICLE_PASS
PRODUCT_TRACKING PAYMENT_RECEIPT GATE_PASS DELIVERY_CONFIRMATION
INSPECTION_TAG }
enum Gender { MALE FEMALE OTHER }
enum MaritalStatus { SINGLE MARRIED DIVORCED WIDOWED }

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// FINANCIAL ENUMS (Production -Grade from comparison reports)

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enum PaymentStatus { PENDING PARTIAL PAID PAID OVERDUE CANCELLED REFUNDED

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enum PaymentStatus { PENDING PARTIAL_PAID PAID OVERDUE CANCELLED REFUNDED
DISPUTED REVERSED FAILED ON_HOLD }
enum PaymentMethod { CASH CHECK BANK_TRANSFER MOBILE_MONEY CREDIT_CARD
DEBIT_CARD CRYPTO CHEQUE STANDING_ORDER DIRECT_DEBIT PROMISSORY_NOTE
POST_DATED_CHEQUE }
enum PaymentCycle { DAILY WEEKLY BIWEEKLY MONTHLY QUARTERLY SEMI_ANNUAL
ANNUAL CUSTOM }
enum RentalStatus { DRAFT ACTIVE SUSPENDED EXPIRED TERMINATED RENEWED
ON_HOLD ARCHIVED }
enum DelinquencyStatus { COMPLIANT WARNING FIRST_NOTICE SECOND_NOTICE
FINAL_NOTICE EVICTION_NOTICE IN_ARBITRATION RESOLVED }
enum DelinquencyLevel { NONE LEVEL_1 LEVEL_2 LEVEL_3 LEVEL_4 LEVEL_5
LEVEL_6 LEVEL_7 }
enum RevenueCategory { RENT MAINTENANCE SERVICE UTILITY PARKING SECURITY
VAT PENALTY OTHER_CHARGES }
enum RefundReason { OVERPAYMENT DUPLICATE SERVICE_ISSUE CANCELLATION ERROR
POLICY DISPUTE_LOST ADJUSTMENT }
enum TransactionType { REVENUE EXPENSE REFUND ADJUSTMENT CORRECTION
TRANSFER WRITEOFF PROVISION }
enum InvoiceStatus { DRAFT SENT VIEWED PARTIALLY_PAID PAID OVERDUE
CANCELLED CREDITED DISPUTED }
enum InvoiceType { STANDARD PROFORMA CREDIT_NOTE DEBIT_NOTE REVISED
RECURRING DEPOSIT FINAL }
enum ReceiptType { RENT_PAYMENT DEPOSIT_RECEIPT ADVANCE_RECEIPT
MAINTENANCE_RECEIPT SERVICE_RECEIPT REFUND_RECEIPT ADJUSTMENT_RECEIPT
CONSOLIDATED }
enum TaxType { VAT GST SALES_TAX PROPERTY_TAX INCOME_TAX CORPORATE_TAX
MUNICIPAL_TAX ENVIRONMENTAL_TAX OTHER }
enum TaxCalculationMethod { PERCENTAGE FIXED_AMOUNT TIERED PROGRESSIVE
FLAT_RATE EXEMPTED ZERO_RATED }
enum DepositType { SECURITY ADVANCE_RENT HOLDING KEY_DEPOSIT EQUIPMENT
UTILITY_DEPOSIT }
enum LatePaymentPenaltyType { PERCENTAGE FIXED_AMOUNT DAILY_RATE
TIERED_PERCENTAGE NONE }
enum ContractTermType { FIXED_TERM PERPETUAL AUTO_RENEWAL ONE_TIME TRIAL
SEASONAL }
enum PaymentScheduleType { ADVANCE ARREARS MID_CYCLE DEPOSIT_ADVANCE
CUSTOM }
enum DiscountType { VOLUME EARLY_PAYMENT LOYALTY SEASONAL REFERRAL BUNDLE
PERMANENT PROMOTIONAL GOODWILL LONG_TERM }
enum BillingFrequency { MONTHLY QUARTERLY SEMI_ANNUAL ANNUAL BI_WEEKLY
WEEKLY DAILY CUSTOM }
enum CurrencyCode { UGX USD EUR GBP KES TZS ZAR NGN GHS XOF }
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enum RevenueType { PRIMARY TAX PENALTY }
enum RevenueStatus { ACCRUED RECOGNIZED COLLECTED WRITTEN_OFF }

// DISASTER & RISK ENUMS
enum RiskType { FLOOD FIRE HEALTH SECURITY OPERATIONAL }
enum RiskLevel { LOW MEDIUM HIGH CRITICAL }
enum IncidentSeverity { MINOR MODERATE MAJOR CATASTROPHIC }
enum ClaimStatus { SUBMITTED UNDER_ASSESSMENT APPROVED REJECTED PAID }
enum EmergencyProtocolType { FIRE FLOOD MEDICAL SECURITY EVACUATION }

// ADDITIONAL MODULE ENUMS
enum InspectionStatus { SCHEDULED IN_PROGRESS COMPLETED FAILED
FOLLOW_UP_REQUIRED }
enum ViolationSeverity { MINOR MODERATE MAJOR CRITICAL }
enum IoTDeviceType { CCTV SENSOR ACCESS_CONTROL ENVIRONMENTAL
GATE_CONTROLLER }
enum AlertSeverity { INFO WARNING ERROR CRITICAL }

```

2. COMPLETE MODELS / TABLES LIST (40+ Tables)

Core Tables (User, Auth, Geo, Market Infrastructure)

Table	Key Columns
User	id, email, passwordHash, phone, mfaEnabled, mfaType, totpSecret, status, lastLogin, loginAttempts, lockUntil, deletedAt
UserProfile	id, userId, firstName, lastName, nationalId, taxIdNumber, gender, dateOfBirth, address, mmisId, bankAccountDetails
VerificationToken	id, token, tokenType, userId, expiresAt, isUsed
UserSession	id, userId, sessionToken, deviceInfo, ipAddress, expiresAt, isActive
Geolocation	id, name, code, country, timezone, currency
District	id, geolocationId, name, code
City	id, districtId, name, code
Market	id, cityId, name, uniqueCode, marketType, address, operatingDays, vatRate, status, totalFacilities
MarketLevel	id, marketId, levelNumber, uniqueCode, name, hasElevator, status

MarketSection	id, marketId, levelId, uniqueCode, name, sectionType, supervisorId
MarketAisle	id, sectionId, uniqueCode, aisleNumber, name, status
Facility	id, marketId, levelId, sectionId, aisleId, uniqueCode, unitNumber, type, monthlyRent, status, occupationStatus, qrCode, lastInspection, inspectionDueDate, depositStatus, lockedDate, lockReason

Stakeholder & Business Tables

Table	Key Columns
Stakeholder	id, userId, stakeholderType, kycStatus, taxComplianceStatus
MarketAuthority	id, stakeholderId, authorityName, registrationNumber
Member	id, stakeholderId, membershipNumber, businessName, taxIdNumber
Vendor	id, stakeholderId, vendorCode, businessName, businessLicenseNumber, taxIdNumber, delinquencyStatus, daysOverdue, isDelinquent
Supplier	id, stakeholderId, supplierCode, businessName, licenseNumber, taxId
Customer	id, stakeholderId, customerCode, loyaltyPoints, totalSpent

Financial & Revenue Tables (Enhanced)

Table	Key Columns
Revenue	id, revenueType, category, facilityId, vendorId, amount, currency, status, incurredDate
Invoice	id, invoiceNumber, type, accountId, vendorId, subtotal, taxAmount, totalAmount, dueDate, status
Transaction	id, transactionReference, type, fromAccountId, toAccountId, invoiceId, amount, paymentMethod, status
RentContract	id, facilityId, tenantId, startDate, endDate, monthlyRent, status
RentPayment	id, contractId, amount, paymentDate, status, lateFee
TaxCollectionLog	id, facilityId, taxType, taxAmount, sourceType (DELIVERY/RENT/SALE), remitted
DelinquencyNotice	id, facilityId, vendorId, noticeType, daysOverdue, amountDue, status
DelinquencyHistory	id, vendorId, level, status, date

Operational Modules

Table	Key Columns
GateEntry	id, marketId, gateId, vendorId, facilityId, entryTime, exitTime, vehicleNumber, tokenId, status
MarketToken	id, code, type, status, facilityId, vendorId, expiresAt
QrCode	id, code, type, data, expiresAt, scanCount
Delivery	id, supplierId, facilityId, deliveryDate, status, verifiedBy
DeliveryItem	id, deliveryId, productId, quantity, unitPrice
Inventory	id, productId, facilityId, quantity, batchNumber, expiryDate
StockMovement	id, productId, fromFacilityId, toFacilityId, quantity, movementType
Product	id, sku, name, category, unitPrice, vatRate, barcode
PosTransaction	id, terminalId, vendorId, totalAmount, paymentMethod, items
Receipt	id, transactionId, receiptNumber, type

Disaster, Risk & Compliance

Table	Key Columns
RiskAssessment	id, marketId, riskType, riskLevel, assessmentDate, findings, mitigationPlan
DisasterIncident	id, marketId, incidentType, severity, reportedAt, estimatedLoss, status
CompensationClaim	id, incidentId, vendorId, claimedAmount, assessedAmount, approvedAmount, status
EmergencyProtocol	id, marketId, type, steps, assignedRoles, isActive
Inspection	id, facilityId, inspectorId, inspectionDate, overallScore, status
Violation	id, inspectionId, type, severity, correctiveAction, status
HealthInspection	id, marketId, facilityId, hygieneScore, safetyScore, complianceScore

IoT, CCTV & Surveillance

Table	Key Columns
IoTDevice	id, marketId, type, name, status, streamUrl, lastSeen

CctvFeed	id, deviceId, name, streamUrl, isLive, isRecording
lotAlert	id, deviceId, alertType, severity, message, status
VideoRecording	id, deviceId, startTime, endTime, filePath, expiresAt

Additional Supporting Tables

- AuditLog, Notification, Document, UserRole, RolePermission, Invitation, HandshakeCode, SupplierBid, PurchaseOrder, SupplierRating, Asset, SupportTicket, LoyaltyTransaction, Complaint, BusinessLicense, MarketRegulation, etc.

✅ **This is the complete consolidated list** (40+ tables, 25+ enums).

Your turn:

Please review and tell me:

1. Which attributes/fields you want to **keep / remove / enhance**?
2. Any business logic alignments (e.g., 90-day lockdown rules, tax collection points, delinquency escalation matrix)?
3. Priority modules for the final Prisma schema + relations + ERD?

Once you reply, I will immediately deliver:

- Full `schema .production .prisma` with all relations
- Mermaid ERD
- PlantUML diagram
- Migration plan & deployment checklist

Ready when you are! 🚀